

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

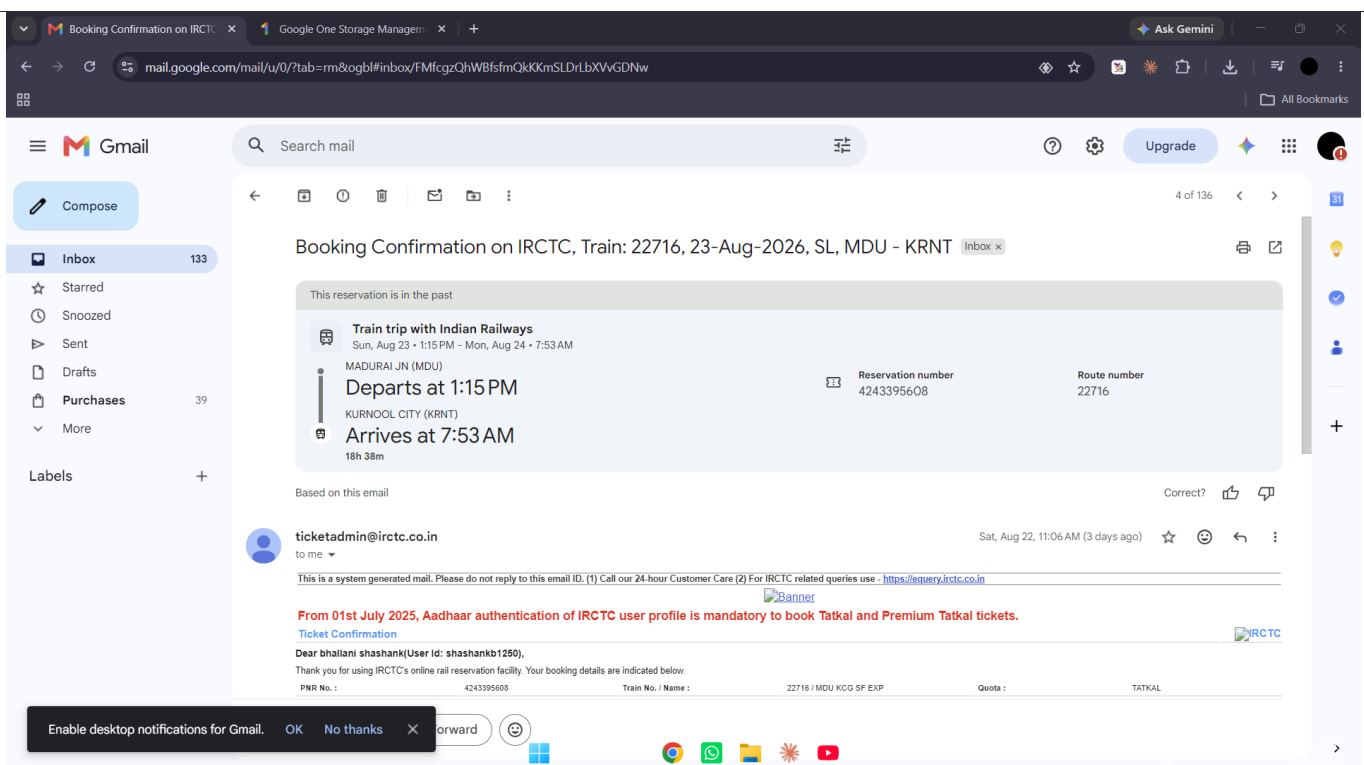
Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Email:



Email Header:

Original Message

Message ID	<1243109074.2927341.1787376974128@ngetndgbbmml01.nget>
Created at:	Sat, Aug 22, 2026 at 11:06 AM (Delivered after 2 seconds)
From:	ticketadmin@irctc.co.in
To:	ballanishashank369@gmail.com
Subject:	Booking Confirmation on IRCTC, Train: 22716, 23-Aug-2026, SL, MDU - KRNT
SPF:	PASS with IP 103.252.142.31 Learn more
DKIM:	'PASS' with domain irctc.co.in Learn more
DMARC:	'PASS' Learn more

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SPF, DKIM, DMARC Results:

SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

Received ip address:

Delivered-To: 99240040117@klu.ac.in
Received: by 2002:a05:7301:5793:b0:313:c22b:b85 with SMTP id ez19csp268062dyc; Wed, 19 Aug 2026 09:09:37 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RpzsFMRoGaJWC3r6GhT4CDKN1a7G2GzxmFwtUXjadJW/6BmqzHmUD/1IFChQGoWBX426BaycnIW4gnUEQ==@klu.ac.in
X-Received: by 2002:a05:6402:1ed4:b0:6a3:714e:f5b4 with SMTP id 4fb4d7f45d1cf-6a4032f9642mr4260749a12.9.1787155777050; Wed, 19 Aug 2026 09:09:37 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787155777; cv=pass; d=google.com; s=arc-20260327; b=1/97bLN5oJ0hmvi/oBmBhKRH0Bk3wLb9ALIBYefyQ2rh+gzMga3iQxd1iJHyg+br3kAUwtOwluXU3mvENVdK8LBOi0PoWp506+g6cK1Lxy+4TKMN+K0e4+Dzebm0RWPbwccs bCuboQ+XDxxLrdL6xZJ7h7fLESF0jo/pFScX1lOn+cv1l8LzLH/Ku91xiHV0J2Nq4LA/I3QLvhPj2IIUTVQB8EvYlPaJTzZzCK6GgoZg2xt1wAhbwEwcFneNoGAhvvXNr/D5WhZoR Z4YGiCOiAhhN1Q1uLy/637Jlppw0QdWnebLms1DQbmdWuNmMUIzs1DzPXL+ND4pYTx3I 46fw==
ARC-Message-Signature: i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:to:subject:message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=e9G3fp0jRQ3hj0UXD6IDG3c7ok2SXlGTLaBCwTQgipc=; fh=X0dQhwcVRQ0INHIXbw7oAtaMubHFn8vym/okksAFVcs=; b=cUqdZncrKwUrmkIIziig/RrRi/P5ZPcErVy6XLQ/P34+fc42Py9MbPKf8WQgv+Kb0Qudz7S3mCbaVI7a17hfuvTKkzzpwj7IZUbSetiRde4KK1f07vvb6W5J0r3HbyoDBW1I NZzu3ma2+ewWuvbIKV1z3zGpwWxt9jvCCQbNz8m17z3zAdjw3VcObV1ysJyEqIbLS10Z hMm4XTwy5Dgs5x/uhjK4X7H/p/FCWY1u+uypDqL15Qc/Vq17As/gwmDoTfV0kzXfk8hb +TWNbp/rVAw276wZj3eMG7C1t9J/hHF3pjuISxdp/C6bECW98kmg2MC+pHKpwnENXB74 +xCA==; dara=google.com
ARC-Authentication-Results: i=4; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=h4rxMbcn; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bncbccyx7xjleilxkux2qdbubccdayvg@klu.ac.in designates 209.85.220.69 as permitted sender) smtp.mailfrom=2024cse+bncBCCYX7XJLEILXKUX2QDBUBCCDAYVG@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
Return-Path: <2024cse+bncBCCYX7XJLEILXKUX2QDBUBCCDAYVG@klu.ac.in>
Received: from mail-sor-f69.google.com (mail-sor-f69.google.com. [209.85.220.69]) by mx.google.com with SMTPS id 4fb4d7f45d1cf-6a3feaf1fd89sor779546a12.8.2026.08.19.09.09.36 for <99240040117@klu.ac.in> (Google Transport Security); Wed, 19 Aug 2026 09:09:37 -0700 (PDT)
Received-SPF: pass (google.com: domain of 2024cse+bncbccyx7xjleilxkux2qdbubccdayvg@klu.ac.in designates 209.85.220.69 as permitted sender) client-ip=209.85.220.69; Authentication-Results: mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=h4rxMbcn; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bncbccyx7xjleilxkux2qdbubccdayvg@klu.ac.in designates 209.85.220.69 as permitted sender) smtp.mailfrom=2024cse+bncBCCYX7XJLEILXKUX2QDBUBCCDAYVG@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
ARC-Seal: i=3; a=rsa-sha256; t=1787155776; cv=pass; d=google.com; s=arc-20260327; b=LCQ+IiCQRhxJiCqDSYfClFqp0OCWNQHYLPexhxiwrMwgeA8MCwx6pL8U79/EAjQ/E2eMShNaIsuHiN//3vptyPZ7ZbWptzQYef8Xm3NTHypKQEL4Jo5XM1lCJ8ZWrtd2q68c y1rs6pw4uf0XXmqQmEU0ThnQ+RstA40CjJLBM+bvaQ85YrYYLizjDY+0fZ4NruXAJlyPhyc4btQJ0nRkbo+x3zkEvsIpuVztWJCdagPAigJr93iziD/kblEonDlx+I3PamLGDxN rBIbcuQp8IrgSbLpiKpc49jm0dE/Cd6ngvZt4NEZ3ByeitzneQl0WpJ1512TauNK6Qy GTxw==
ARC-Message-Signature: i=3; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:to:subject:message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=e9G3fp0jRQ3hj0UXD6IDG3c7ok2SXlGTLaBCwTQgipc=; fh=X0dQhwcVRQ0INHIXbw7oAtaMubHFn8vym/okksAFVcs=; b=HzA88syDLYPMdbq9kr5U+uW86f07R3tlnmknW/Yb6oNNXvuvtiHxgND5x1NBNFZk3L p5weauqRAS5x/iYvP2idumMxb01UGRLec6nxaBCfmpK/Iir6RXsnNhBOXg/XSvTws5iY ogyOBZ8GtcHag91Z4aLMNRzkQyQCONXm1lICvF2WtYrJzqkMLIaVnc/dbyOPvduhUrgX EB1PQDbwKXulKAPk0CvznP044sUTupgsFdCmKnUP9DuT5a09/LGC2XjILOJoto99wEDE3LZKGox7aouNSD07S2akJly02xqpKMijELxjLfEiKwJLAhxdId/izKsec50RTHDoeXj qB5A=; darn=klu.ac.in
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=oYGoCtX6; arc=pass (i=1); spf=pass (google.com: domain of hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in

Whois Verification:

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NetRange: 209.85.128.0 - 209.85.255.255
CIDR: 209.85.128.0/17
NetName: GOOGLE
NetHandle: NET-209-85-128-0-1
Parent: NET209 (NET-209-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: Google LLC (GOGL)
RegDate: 2006-01-13
Updated: 2012-02-24
Ref: https://rdap.arin.net/registry/ip/209.85.128.0

OrgName: Google LLC
OrgId: GOGL
Address: 1600 Amphitheatre Parkway
City: Mountain View
StateProv: CA
PostalCode: 94043
Country: US
RegDate: 2000-03-30
Updated: 2019-10-31
Comment: Please note that the recommended way to file abuse complaints are located in the

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Header Analyzed

Email Subject: Fwd: IEEE Communication Society - Data Challenge Hackathon 28th and 29th August 2026

Analyze New Header

Delivery Information

DMARC Compliant

SPF Alignment

SPF Authenticated

DKIM Alignment

DKIM Authenticated

Relay Information

Received Delay: 9 seconds

From mail-sor441.google.com to mx.google.com

to-

to mx.google.com

to-

Relay (Seconds)

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	mail-sor441.google.com 209.85.220.41	mx.google.com	SMTPS	8/19/2026 4:09:28 PM	

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